RAJALAKSHMI ENGINEERING COLLEGE
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CS23532 – COMPUTER NETWORKS

EX 1: Study of Various Network Commands used in Linux and Windows

AIM :

To study and execute the various network commands used in Linux and Windows and to understand their purpose through hands-on practice.

PROCEDURE :

Windows commands (Command Prompt / PowerShell)

1. arp -a : Address Resolution Protocol. Shows the IP address and MAC address of the devices (such as the router) known to your computer.

2. hostname : The simplest TCP/IP command. Displays the name of your computer.

3. ipconfig /all : Displays detailed configuration information about the TCP/IP connection, including gateway, DNS, DHCP and the type of Ethernet adapter.

4. nbtstat -a : Helps solve problems with NetBIOS name resolution (NBT stands for NetBIOS over TCP/IP).

5. netstat : Network statistics. Displays the active TCP/IP connections of the computer and is used to monitor incoming and outgoing connections, routing tables and interface statistics (e.g. netstat -r).

6. nslookup : Name server lookup. Performs DNS lookups to display the IP address of a computer, the MX records or the name servers of a domain (e.g. nslookup www.google.com).

7. pathping : Unique to Windows; a combination of ping and tracert. It traces the route to the destination and then tests each router along the way to gather statistics on packet loss at each hop.

8. ping : Packet INternet Groper. Tests connectivity between two nodes using ICMP. It can be used with a hostname (ping localhost), an IP address (ping 4.2.2.2) or a domain name (ping www.facebook.com).

9. route print : Shows and manipulates the IP routing table; mainly used to set up static routes to specific hosts or networks.

Linux commands

1. ip : Shows address information, manipulates routing and displays network devices, interfaces and tunnels.

```
ip address show
ip address add 192.168.1.254/24 dev enps03
ip link set eth0 up
```

```
ip route add default via 192.168.1.254 dev eth0
ip route get 10.10.1.4
```

2. ifconfig : Configures and troubleshoots network interfaces; it has since been replaced by the ip command.

```
ifconfig
```

3. mtr : Matt's traceroute. Combines the functions of ping and traceroute and shows statistics such as response time and packet loss for every hop.

```
mtr google.com
mtr -c 10 google.com
```

4. tcpdump : Captures and displays network packets.

```
tcpdump -D
tcpdump -i eth0 -c 10
tcpdump -i eth0 host 8.8.8.8
tcpdump -i eth0 port 53
```

5. ping : Verifies IP-level connectivity to another TCP/IP computer by sending ICMP Echo Request messages and showing the round-trip times.

```
ping google.com
ping -c 10 google.com
```

6. nmcli : Manages the network connection settings of an Ethernet connection from the command line.

```
nmcli connection show
nmcli connection modify "Wired connection 1" ipv4.method auto
```

Steps

1. Open Command Prompt or PowerShell on the Windows system.
2. Type each of the Windows commands listed above and press Enter.
3. Observe and record the output of each command.
4. Open a terminal on a Linux system and execute the Linux commands listed above.

OUTPUT:

arp -a, hostname and ipconfig /all

```
PS C:\Users\hemes> arp -a

Interface: 10.87.222.52 --- 0x3
  Internet Address      Physical Address      Type
  10.87.222.194         3a-10-b6-30-a4-0b     dynamic
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0x8
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
PS C:\Users\hemes> hostname
HemeshKumar31
PS C:\Users\hemes> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : HemeshKumar31
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-08
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::e0ec:779a:3240:ff27%8(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

nbtstat -a, netstat and nslookup

```
PS C:\Users\hemes> nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.

PS C:\Users\hemes> netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.87.222.52:49410     4.213.25.241:https     ESTABLISHED
  TCP    10.87.222.52:49957     49.44.131.96:https     ESTABLISHED
  TCP    10.87.222.52:50347     150.171.27.11:https    TIME_WAIT
PS C:\Users\hemes> nslookup www.google.com
Server:  UnKnown
Address:  10.87.222.194

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482b:7700::
```

nslookup, pathping and ping

```
PS C:\Users\hemes> nslookup www.google.com
Server:  UnKnown
Address:  10.87.222.194

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482b:7700::
          2001:4860:482d:7700::
          2001:4860:482a:7700::
          2001:4860:482c:7700::
          2001:4860:4828:7700::
          2001:4860:4826:7700::
          2001:4860:4827:7700::
          2001:4860:4829:7700::
          142.251.157.119
          142.251.150.119
          142.251.155.119
          142.251.156.119
          142.251.152.119
          142.251.154.119
          142.251.153.119
          142.251.151.119

PS C:\Users\hemes> pathping google.com

Tracing route to google.com [142.251.43.46]
over a maximum of 30 hops:
  0  HemeshKumar31 [10.87.222.52]
  1  10.87.222.194
  2  192.0.0.1
  3     *        *        *
Computing statistics for 50 seconds...
PS C:\Users\hemes> ping localhost ping 8.8.8.8
Bad parameter ping.
PS C:\Users\hemes> ping localhost

Pinging HemeshKumar31 [::1] with 32 bytes of data:
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
```

ping and route print

```
PS C:\Users\hemes> ping localhost

Pinging HemeshKumar31 [::1] with 32 bytes of data:
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms

Ping statistics for ::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\hemes> route print
===========================================================================
Interface List
  8...0a 00 27 00 00 08 ......VirtualBox Host-Only Ethernet Adapter
 20...00 ff eb 46 7f 58 ......TAP-Windows Adapter V9
  4...c2 35 32 6d bf e1 ......Microsoft Wi-Fi Direct Virtual Adapter
 17...c6 35 32 6d bf e1 ......Microsoft Wi-Fi Direct Virtual Adapter #2
  3...c0 35 32 6d bf e1 ......Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0    10.87.222.194     10.87.222.52     35
      10.87.222.0    255.255.255.0         On-link      10.87.222.52    291
     10.87.222.52  255.255.255.255         On-link      10.87.222.52    291
    10.87.222.255  255.255.255.255         On-link      10.87.222.52    291
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
     192.168.56.0    255.255.255.0         On-link      192.168.56.1    281
     192.168.56.1  255.255.255.255         On-link      192.168.56.1    281
   192.168.56.255  255.255.255.255         On-link      192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link      192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link      10.87.222.52    291
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
```

RESULT:

Thus, the various network commands used in Linux and Windows were studied and executed successfully, and their outputs were observed.